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A framework for bias-aware dataset evaluation in soft facial attribute recognition

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ABSTRACT

Soft Facial Attribute Recognition (FAR) remains largely unexplored in terms of demographic fairness. To the best of our knowledge, this study presents one of the first comprehensive analyses of demographic bias in FAR, proposing a systematic framework to detect, quantify, and promote awareness of both representational and stereotypical biases, supporting their mitigation. Leveraging established taxonomies, we evaluate state-of-the-art datasets using a rigorous set of interpretable bias metrics to uncover hidden demographic imbalances. To support reliable fairness assessment, we first enrich the datasets with standardized demographic annotations using the FairFace model. We then address label inconsistencies through the integration of predictions from advanced Vision-Language Models (VLMs). Our analysis reveals substantial imbalances across gender, age, and racial categories-specifically White, Black, and Asian- affecting dataset composition. Furthermore, we show that conventional fairness metrics often yield divergent assessments, highlighting the importance of multi-metric evaluation. This study provides a replicable methodology and actionable insights to support bias-aware facial analysis.

1. Introduction

Facial Attribute Recognition (FAR) is a core task in computer vision focused on predicting interpretable facial characteristics, such as *Big Lips*, or *Gray Hair*. These soft biometrics provide semantically rich information useful for recognition, verification, and image retrieval, particularly in unconstrained settings [9]. Although not sufficient for unique identification, soft attributes are widely used in surveillance and biometric systems to enhance performance in real-world conditions [32]. Despite their broad adoption, FAR systems have received limited attention in terms of demographic fairness, defined as consistent performance across population groups. This issue is especially critical given that demographic traits, though rarely annotated, strongly influence facial appearance. The lack of standardized demographic labels and subjective annotations, makes FAR vulnerable to two major forms of bias: *representational bias*, which manifests as systematic underrepresentation of certain demographic groups, and *stereotypical bias*, characterized by problematic associations between specific attributes and demographic categories. Studies show that facial recognition systems often underperform on individuals with darker skin tones and systematically favor the dominant demographic groups represented in their training data (e.g., Caucasians in Western-trained models, East Asians in models developed in Asia). These disparities reflect limited demographic diversity in training sets and raise critical concerns about fairness and reliability in AI systems [20]. Deep learning models often amplify biases embedded in training data, including harmful social stereotypes [28]. These biases can carry through to predictions, reinforcing existing disparities. Tackling this issue requires assessing not only model outputs but also the data itself-an often overlooked step that is essential for understanding bias propagation, validating mitigation strategies, and identifying the demographic contexts in which models can be trusted. A deeper understanding of structural bias in datasets is therefore essential. Popular facial datasets like CelebA [18] and Labeled Faces in the Wild (LFW) [10] have been central to training and evaluating FAR models, but both exhibit structural limitations that reinforce demographic bias. CelebA, widely used for binary attribute classification [21], shows severe class imbalance-non-blond individuals outnumber blonds by 5.7 to 1, and blond males make up only 0.86 % of the training set [26]-leading to systematic misclassification of underrepresented groups. In addition, CelebA contains subjective and culturally biased labels-such as

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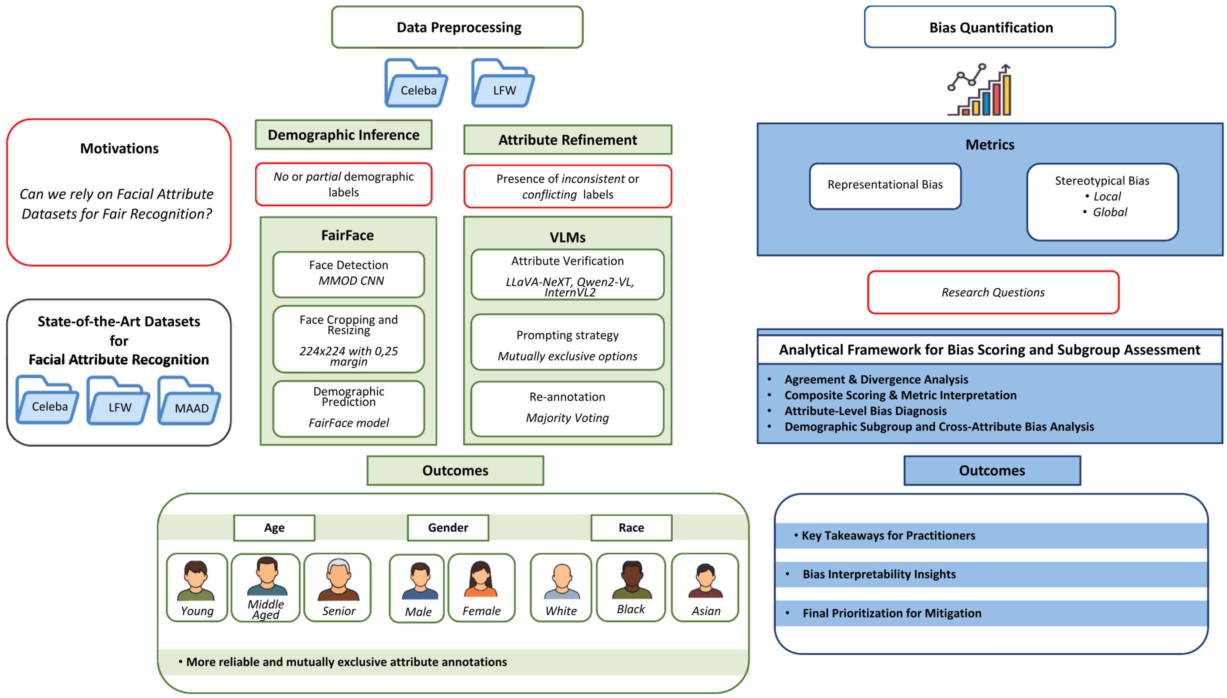

Fig. 1. Overview of the proposed framework for guiding bias mitigation in soft FAR. It standardizes preprocessing (FairFace, VLMs), quantifies representational and stereotypical bias, and supports validation, scoring, and intersectional analysis-enabling fairness-aware model development. Core motivations are marked in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Attractive and *Wearing Lipstick*-which reflect Western beauty standards, often skewing predictions toward white, blond women and amplifying such biases during training [16,30]. Similarly, LFW overrepresents light-skinned, Western individuals and underrepresents those from the Global South, limiting its value for fairness evaluations [25]. Identity overlaps between training and test sets further compromise the reliability of subgroup analyses. Overall, these structural flaws reflect three key forms of bias: *biological bias* (imbalanced representation by age, sex, or ethnicity), *cultural bias* (influenced by label choices and aesthetic norms), and *technical bias* (due to class imbalance, sparse metadata, or flawed dataset design). This taxonomy provides a practical lens to assess dataset limitations and support the creation of more inclusive FAR benchmarks.

1.1. Motivations and contributions

While interest in Soft FAR has grown with the rise of deep learning, its demographic biases remain largely underexplored-especially when compared to related areas like face recognition and expression analysis [8,33]. To fill this gap, we present, to the best of our knowledge, the first systematic dataset-level investigation of demographic bias in soft FAR Fig. 1. Our framework combines standardized dataset evaluation with refined annotations to uncover both representational and stereotypical biases-modeling the latter through attribute-demographic pairs- to support the implementation of data-driven mitigation approaches. Our key contributions are:

- **Systematic assessment of bias in soft attribute prediction.** We conduct a rigorous evaluation of three selected FAR datasets, extracting demographic profiles and applying a taxonomy-aligned set of bias metrics to uncover disparities across demographic groups and dataset sources.
- **Definition of minimal and interpretable bias metrics.** We define a compact, interpretable set of bias metrics that effectively summarize demographic biases in FAR datasets, supporting structured evaluation and mitigation strategies.

Beyond the core contributions, we provide additional analyses to improve dataset quality and fairness assessment:

- **Standardized demographic annotations for LFW and CelebA.** Using the FairFace model, we provide consistent demographic labels for these popular datasets, filling a key gap in fairness assessment and enabling more reliable bias evaluation.
- **Refinement of soft attribute annotations.** Through the use of three state-of-the-art VLMs, we identify and resolve inconsistencies in existing soft attribute labels, improving their reliability and consistency across demographic groups.

The remainder of this paper is organized as follows: Section 2 reviews FAR methods; Section 3 presents bias metrics; Section 4 details experimental setup including dataset preprocessing and research questions; Section 5 analyzes representational and stereotypical bias patterns; Section 6 concludes with future directions.

2. Literature review

Several works on bias detection [14] and mitigation [23] have highlighted the role of soft biometric attributes as hidden proxies for protected characteristics like gender or race, potentially reinforcing unfair model behavior. Ramaswamy et al. [24] propose a GAN-based latent space perturbation method that generates synthetic image pairs sharing the same target attribute but differing in the protected attribute. This strategy reduces spurious correlations and improves fairness metrics on CelebA, particularly for gender-independent soft attributes. However, the approach relies on linear separability assumptions in the latent space and is less effective on noisy or gender-sensitive labels. Ozturk et al. [22] focus on facial hairstyle as a soft attribute influencing face recognition performance. Their study shows that even with balanced training data or localized augmentation, significant accuracy gaps persist across facial hair conditions. These effects are also demographic-dependent, revealing that soft attributes can act as confounding factors in fairness assessments.

Despite these advances, a systematic framework to quantify the contribution of soft attributes to representational and stereotypical bias across demographic groups is still lacking. Existing approaches typically focus on isolated cases or predefined protected attributes, without analyzing how soft attribute predictions interact with group distributions at scale.

3. Metrics for dataset bias evaluation

We adopt the taxonomy from [8], which distinguishes *representational* from *stereotypical* bias. Key metrics and their interpretation are outlined below; formal definitions are provided in the Supplementary Material (Tables S1-S2).

Representational bias. Representational bias is assessed through four categories of metrics, each reflecting a different aspect of group distribution. *Richness* metrics count the number of distinct groups, regardless of frequency. *Evenness* metrics evaluate how uniformly samples are distributed across groups. *Dominance* metrics highlight cases where one group is overrepresented. *Combined* metrics integrate multiple aspects-typically richness and evenness-into a single value. Together, these perspectives should provide a comprehensive view of a dataset's demographic structure. We start with **Richness** $R(X)$, which counts the number of distinct demographic groups. Although simple to compute, it ignores group sizes and fails to detect distributional imbalances. Moreover, it does not indicate how many groups should be expected: low richness may be acceptable in binary contexts but insufficient when multiple subpopulations matter. To address this, we consider metrics of distributional evenness. The **Shannon Evenness Index** (SEI) captures how uniformly samples are spread across observed groups. Normalized by the number of present groups, it enables comparisons across datasets with different group counts and is robust to small variations. However, it ignores unobserved but potentially relevant groups, meaning a high SEI may still indicate balance over a narrow subset. Moreover, SEI is undefined in single-group scenarios. A complementary metric is the **Normalized Standard Deviation complement** (1-NSD), which assesses evenness based on variance. It captures deviations from perfect uniformity and is particularly sensitive to large disproportions. Unlike entropy-based measures, it directly reflects dispersion, but shares similar limitations: it is undefined for single-group cases, ignores unobserved groups, and may be unstable in near-uniform distributions. Among dominance metrics, the **Inverted Imbalance Ratio** ($IR^{-1}(X)$) compares the least and most represented groups, while the complement of the **Berger-Parker Index** (1-BP) reflects the dominance of the largest group. Both are simple and intuitive, but focus on extremes, potentially missing subtler imbalances. Within combined metrics, **Shannon Entropy** ($H(X)$) captures the uncertainty in group membership by accounting for both group count and frequency. The **Effective Number of Species** (ENS) translates this into an interpretable quantity: the number of equally represented groups needed to produce the same entropy. Simpson-based metrics-including **Simpson's Index** (D), its complement (1-D), and reciprocal ($1/D$)-capture both richness and distribution. The reciprocal form is especially interpretable, representing the effective number of equally sized groups needed to match the observed diversity. While conceptually similar to ENS, it is derived from a distinct formulation.

Stereotypical bias. We distinguish between *global* and *local* analyses. Global analysis measures the overall dependence between a group attribute and the target variable, providing an aggregate association score (e.g., how hair color distributions vary by gender). Local analysis instead focuses on specific group-class pairs, highlighting over- or under-representations relative to an expected baseline (e.g., whether women are disproportionately labeled as *Blonde Hair*). To quantify global stereotypical bias, we adopt association metrics derived from the chi-squared statistic: **Cramér's V** (ϕ_C), **Tschuprow's T** (T), and the **Contingency Coefficient** (C). These metrics normalize the Pearson chi-squared value to a $[0, 1]$ scale, where higher values indicate stronger dependence between group and target variables. While they share the same range, they differ in normalization: ϕ_C is the most widely used, as it ensures comparability across asymmetric tables; T is more conservative and suited for symmetric tables; C is bounded above by a table-dependent maximum, limiting its interpretability. Interpretation thresholds for ϕ_C depend on the number of degrees of freedom (DoF), which in turn reflect the size of the contingency table. For DoF $= 1$, *small*, *medium*, and *large* bias correspond to values of 0.1, 0.3, and 0.5, respectively; values ≥ 0.5 are considered *strong*. For DoF >1, thresholds scale inversely with DoF. In addition to chi-squared-based metrics, we include association measures from information theory. The **Theil's Uncertainty Coefficient** (U) quantifies the proportion of uncertainty in the target explained by the group attribute. Being asymmetric, it is useful when the direction of dependence matters. We use $U^{\rightarrow}$ for group-to-target association, and $U^{\leftarrow}$ for the reverse. Local metrics provide a fine-grained view by evaluating specific group-class combinations. The **Normalized Pointwise Mutual Information** ($NPMI$) measures deviation from independence on a ($[-1, 1]$) scale, with higher absolute values indicating stronger associations. **Ducher's Z** (Z) refines this notion by scaling deviations based on the maximum possible divergence implied by marginal frequencies, improving comparability under class imbalance. **Lewontin's D** (D_L) computes the difference between observed and expected co-occurrence probabilities. Although unnormalized, it is interpretable and effective when absolute deviation is of interest. *Observation*: Both D_L and Z are antisymmetric under negation when the group attribute is binary.

4. Experimental analysis: demographic bias in facial attribute recognition

This section examines demographic bias in facial attribute datasets by analyzing representational imbalances and stereotypical associations between demographic groups and soft-biometric traits. The dimensions-**Gender** (*Male, Female*), **Age group** (*Young, Middle-Aged, Senior*), and **Race** (*White, Black, Asian*)-are defined according to the MAAD benchmark taxonomy [27], allowing for standardized fairness evaluation and cross-dataset comparison.

4.1. Datasets

CelebA dataset [18] contains 202,599 images of 10,177 celebrities, with an average of 20 images per identity. Each image is annotated with 40 binary soft-biometric attributes (e.g., gender, baldness, gray hair, lipstick). CelebA captures wide intra-subject variability across ethnicity, age, pose, expression, occlusion, and lighting, making it a reference benchmark for facial attribute recognition and related tasks. However, the dataset lacks transparency regarding annotation methodology, as labels were assigned by a commercial service using undisclosed procedures.

LFW dataset [10] includes 13,200 images of 5700 individuals captured in unconstrained conditions. It offers 74 binary attributes spanning demographic traits, hair/skin features, and environmental properties. Unlike CelebA, LFW attributes are derived from classifier prediction scores rather than manual annotations, producing continuous values. Prior studies have shown that many of these scores cluster near zero, indicating low confidence and unreliable annotations, with only 72 % agreement with human-labeled ground truth [27]. To improve reliability, we apply an empirically validated discretization: values below 0 are mapped to -1 (absent), values between 0 and 0.3 to 0 (uncertain), and values above 0.3 to 1 (present).

MAAD [27] is a large-scale facial dataset built on VGGFace2 [4], comprising 3.3 million images from over 9000 individuals. It provides 123.9 million binary annotations for 47 soft-biometric attributes, generated via a transfer-learning pipeline that maps labels from multiple source datasets. Compared to CelebA and LFW, MAAD offers broader coverage and improved label consistency, validated by a panel of human annotators. Attributes are encoded as 1 (present), -1 (absent), and 0 (undefined), allowing fine-grained demographic analysis.

Notably, there is no identity overlap between the three datasets, ensuring that cross-dataset comparisons are not affected by identity leakage. For our experimental analysis, we select attributes from CelebA, including composite categories such as *Hair Color*, defined as a mutually exclusive group of binary labels (e.g., *Black Hair*, *Blond Hair*, *Brown Hair*, *Gray Hair*). Although annotated independently, these attributes are grouped under a unified semantic category for analysis. Additional details on dataset composition, selected attributes, and their distribution across gender, age, and race are provided in the Supplementary Material.

4.2. Data preprocessing

Missing demographic labels are inferred via FairFace, while inconsistent attribute annotations are corrected using an ensemble of VLMs. Despite their state-of-the-art performance, both may introduce minor errors or reflect existing biases, especially in ambiguous cases.

4.2.1. Inferring demographic labels: FairFace

An in-depth analysis of demographic bias in soft-facial attribute datasets requires access to demographic information about the depicted individuals. However, most datasets either lack such metadata or provide it only in limited form. Among those considered, only MAAD includes explicit annotations related to *Age group* and *Race*, increasing its relevance for fairness and bias detection. To address the lack of demographic labels in CelebA and LFW-and following the methodology in [8]-bias is analyzed through inferred demographic predictions using FairFace [12], a secondary model trained on 108,501 facial images from Flickr. FairFace offers a diverse label set, including seven ethnic categories, binary *Gender,* and nine *Age* groups. Labels are assigned by external annotators, capturing apparent rather than self-declared characteristics. Demographic inference was performed using the original FairFace pipeline, based on a Max-Margin Object Detection CNN from DLIB. Detected faces were standardized to 224×224 pixels with a 0.25 margin, and only the first face (face0) was analyzed, consistent with the FairFace protocol. Predictions were applied to the entire CelebA and LFW datasets, enriching existing annotations-particularly for *Gender* and *Race*. A consistency check revealed less than 2 % discrepancy in gender labels across datasets. For compatibility with MAAD, we adopted the FairFace *race4* grouping-White, Black, Asian, and Indian-excluding the Indian category from the analysis. Age was mapped into three groups: *Young* (0–29), *Middle-Aged* (30–59), and *Senior* (60 +), yielding a coherent demographic framework. The resulting annotations have been released for public use and benchmarking.

4.2.2. Facial soft attribute relabelling: vision-language models

Accurate annotation of soft facial attributes in datasets remains a complex challenge, often affected by errors and inconsistencies. Prior studies on CelebA and LFW [31] have documented overlaps between mutually exclusive attributes-such as subjects labeled with both *No Beard* and *Goatee*-as well as conflicts in *Hair Color* and *Baldness*, where incompatible traits were simultaneously assigned. To address these issues, we conducted a systematic relabeling process using a majority-voting scheme across three state-of-the-art Vision-Language Models (VLMs): **LLaVA-NeXT** [17], **Qwen2-VL** [29], and **InternVL2.5** [7]. Recent literature highlights increasing interest in VLMs not only for general vision-language understanding, but also for tasks such as image captioning, region grounding, and attribute prediction across domains including medical imaging, fashion analysis, and surveillance [11,13]. Furthermore, Large Language Models (LLMs) have been integrated with vision systems for multimodal tasks, as surveyed in [6], and have proven effective in contexts such as Visual Question Answering (VQA) [2], where linguistic and visual reasoning are combined. Most existing studies have focused on general-purpose annotation or on the perceptual effects of facial attribute editing, rather than on systematic relabeling strategies explicitly designed to mitigate bias [3,19]. Our proposed framework exploits VLM agreement to correct inconsistent annotations, guided by semantic exclusivity constraints. Final labels are assigned through majority voting, with a label accepted if predicted by at least two out of three models. In the rare cases where no consensus is reached (e.g., 0.34 % of *Hair Color* labels in CelebA), manual correction is applied to preserve annotation integrity. Importantly, the revised annotations are not regarded as end-goals but as essential inputs for downstream fairness evaluation pipelines, a context in which label consistency is crucial yet frequently neglected. Model outputs were standardized using predefined textual prompts (see Supplementary Material) and normalized to avoid ambiguity (e.g., replacing *White* with *Gray* in *Hair Color)*. The goal of relabeling is not to redefine ground truth, but to produce internally coherent annotations that support reliable demographic fairness evaluation. All annotations were generated using the following model versions: `Qwen2-VL-7B-Instruct`, `InternVL2_5-2B`, and `llama3-llava-next-8b-hf`.

4.3. Measuring bias in attribute annotations

Following the preprocessing phase, two refined datasets for CelebA and LFW were obtained, where inconsistencies in attribute annotations were resolved and the necessary demographic labels were integrated for subsequent bias analysis. To ensure a more reliable evaluation, ambiguous or neutral annotations were systematically removed from all three datasets. A comprehensive bias analysis was then conducted, assessing both representational bias, by examining the distribution of attributes to identify potential imbalances, and stereotypical bias, by analyzing correlations between facial attributes and demographic categories. The evaluation was performed by computing the metrics presented in Section 3.

4.4. Statistical agreement and divergence analysis of bias metrics

To evaluate the consistency and comparability of bias metrics, a unified analytical framework was introduced, jointly assessing both ordinal and numerical agreement across metric outputs. All scores were standardized via *Z-score normalization*, enabling meaningful cross-metric comparisons by centering values around zero and scaling to unit variance. Ordinal agreement was measured using *Kendall's Tau*, a non-parametric statistic that evaluates rank correlation through concordant and discordant pairs. Compared to Spearman's rho, it offers greater robustness to ties and skewed distributions, and yields more efficient estimates with small to moderate sample sizes. Numerical concordance was assessed through the *Concordance Correlation Coefficient (CCC)* [5,15], which jointly considers correlation and deviation from the identity line (i.e., perfect agreement). The *U-statistics* method was employed for CCC estimation, offering a distribution-free approach suitable for non-Gaussian or heterogeneous data. Lastly, to assess whether pairs of bias metrics could be considered practically equivalent, we employed the *Two One-Sided Tests* (TOST) procedure. This approach evaluates whether the observed difference between two metrics lies within a predefined equivalence margin Δ, representing the maximum discrepancy considered negligible for practical purposes. The equivalence margin Δ was set pragmatically at 20 % of the standard deviation of the differences. Although no established guidelines exist for bias metrics, this threshold reflects a conservative stance-treating differences smaller than typical variability as practically irrelevant. In cases of low dispersion (i.e., when the standard deviation was very small), a fallback margin equal to 5 % of the mean difference was used, with a hard lower bound of $\Delta = 0.01$ to ensure numerical stability.

4.5. Research questions

To guide our analysis, we pose six research questions grouped into two axes: inter-metric consistency and dimensionality (RQ1-RQ2), and attribute-level bias patterns and their demographic interpretability (RQ3-RQ6):

- **RQ1.** Which representational bias metrics yield the highest agreement in capturing attribute-level imbalance, and can a reduced set of metrics be identified without significant informational loss?
- **RQ2.** Do composite bias scores derived from metric aggregation reliably identify highly imbalanced attributes, and how robust are these scores across datasets?
- **RQ3.** Which soft visual attributes consistently exhibit strong representational or stereotypical biases across demographic groups, and how do such patterns vary across datasets?
- **RQ4.** How do global and local stereotypical bias metrics align or diverge in highlighting biased attribute-demographic associations, and what does this divergence reveal about the multifaceted nature of bias?
- **RQ5.** What role does directional asymmetry play in exposing different forms of demographic influence (e.g., leakage vs. imprinting), and how stable are these patterns across datasets?
- **RQ6.** Which attribute-demographic pairs show the highest inter-metric disagreement, and how can this variability support the development of more robust bias auditing strategies?

5. Results and discussion

This section reports the main findings of our bias evaluation framework, emphasizing metric-level agreement and dataset-level patterns. We examine the consistency among bias metrics and identify key demographic and attribute-related trends across datasets, providing insights into metric reliability and the structure of visual biases in facial attribute datasets.

5.1. Representational bias analysis

5.1.1. Agreement bias metrics

Given the heterogeneous nature and scale of the computed representational bias metrics, we first assessed their ordinal concordance using Kendall's Tau rank correlation. The metric R was excluded due to its discrete encoding, which is not directly comparable to the continuous-valued metrics. We found strong positive correlations among several metrics-particularly 1-D, $1/D$, 1-NSD, and ENS-with Kendall's Tau values exceeding 0.90 in most cases. As expected, D was perfectly negatively correlated with $1/D$, while metrics such as SEI and IR^{-1} showed slightly lower but still substantial correlations. The full correlation matrix is reported in Fig.S1 (Supplementary Material). To further investigate this redundancy and improve the interpretability of the metrics, we applied Principal Component Analysis (PCA) to identify a reduced subset that preserves most of the variance. This analysis allowed us to select metrics that contribute meaningfully to the main components while exhibiting limited overlap. The resulting subset provides a more compact representation of the metric space, retaining essential information with reduced redundancy.

5.1.2. Composite attribute ranking and agreement across metrics

To assess representational disparities across visual attributes, we computed composite scores by averaging standardized (Z-scored) values across all continuous bias metrics, excluding R. All metrics were aligned such that higher values consistently indicate lower representational bias. Fig. 2 reports the average composite score for each attribute, aggregated over CelebA, LFW, and MAAD. Attributes such as *Mustache*, *Goatee*, and *Bald* consistently exhibit pronounced disparities, reflected by markedly negative composite scores. In contrast, features like *Age*, *Hair Color*, and *Big Lips* appear comparatively more balanced. However, higher composite scores should not be interpreted as evidence of neutrality, as intersectional or intra-group imbalances may still persist. To evaluate the internal consistency of the composite Z-score as a summary indicator, we computed Kendall's Tau rank correlations between each individual metric and the composite ranking. Metrics focused on dominance and combined spread achieved near-perfect concordance ($\tau > 0.98$), while evenness-based metrics such as 1-NSD and SEI also showed strong agreement (τ in the [0.87–0.90] range). These results confirm that the composite ranking offers a robust and representative synthesis of the underlying metrics.

5.1.3. Metric redundancy analysis and dimensionality reduction

The high concordance between individual metrics (see Supplementary Fig. S1) suggests a substantial degree of redundancy among the bias measures. To formally assess this, we conducted a Principal Component Analysis (PCA) on the standardized metric space. The results confirmed this intuition: the first principal component (PC1) alone accounted for 99.0 % of the total variance, with PC2 and PC3 contributing marginally (0.9 % and 0.1 %, respectively). This indicates that the full metric set is effectively governed by a single latent dimension, capturing highly overlapping statistical signals. Among the original metrics, all contributed almost equally to PC1, but the metric $1/D$ emerged as the most dominant loading (loading = 0.678), suggesting it encapsulates much of the shared variance across the full set. To address the observed redundancy among representational bias metrics, we conducted a systematic metric selection process. We first evaluated a *conceptually diverse* subset comprising four metrics designed to capture distinct dimensions of bias: D (dominance), H (entropy), SEI (normalized entropy), and IR^{-1} (min–max ratio). Despite their theoretical differences, these metrics were found to be highly correlated (mean $r = 0.979$), with the first principal component (PC1) alone explaining 98.5 % of the total variance. Subsequently, we applied a *numerical optimization* strategy to identify the four metrics with the lowest average pairwise correlation: 1-BP, H, IR^{-1}, and SEI. Even this mathematically optimized subset showed only a marginal improvement in variance dispersion, with PC1 accounting for 98.4 % and PC2 for 1.6 %. The persistence of extreme unidimensionality across both conceptually and numerically optimized subsets confirms that representational bias in our dataset manifests as a singular, pervasive latent factor. Rather than suggesting a methodological shortcoming, this result highlights the robustness and convergence of diverse bias metrics toward a common underlying structure. To validate the practical impact of metric reduction, we compared composite Z-score rankings derived from the full nine-metric set against those computed using the optimized four-metric subset. Scores were recalculated for all visual attributes across the three datasets (CelebA, LFW, MAAD) under both configurations. The resulting rankings exhibited strong concordance, with most attributes preserving identical or near-identical positions. A paired-sample t-test conducted over all attributes confirmed that the differences between the full and reduced sets were not statistically significant ($t = -0.829$, $p = 0.415$). The attribute *Hair Color* showed the largest deviation, with a score decrease of $\Delta = 0.27$ attributable to the exclusion of the *ENS* metric. This instability persisted even after expanding the reduced set to include *ENS* as a fifth metric, suggesting that the variability stems from the attribute's intrinsic representational complexity rather than from metric selection artifacts. These findings point to a broader methodological implication: attributes with complex or highly granular categorical structures may be inherently more sensitive to metric formulation. In light of these findings, we advocate for the use of compact, non-redundant, and interpretable metric subsets in future bias auditing pipelines. The selected four-metric configuration strikes a balance between descriptive power

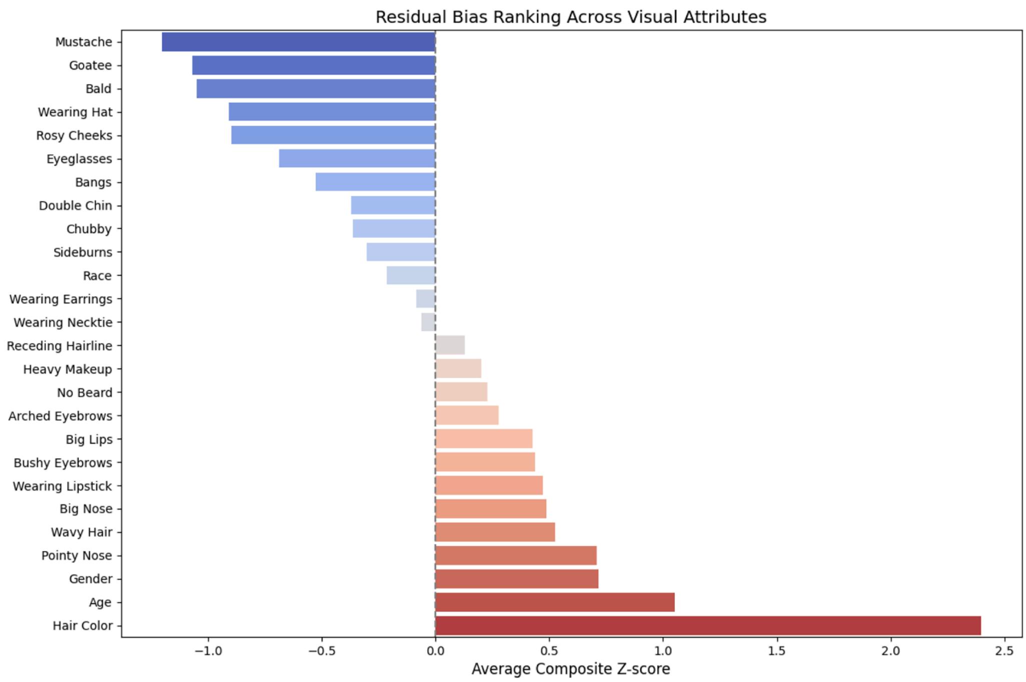

Fig. 2. Composite *Z*-scores averaged across bias metrics for each soft attribute. Positive values indicate balanced representation; negative values highlight systematic demographic imbalance.

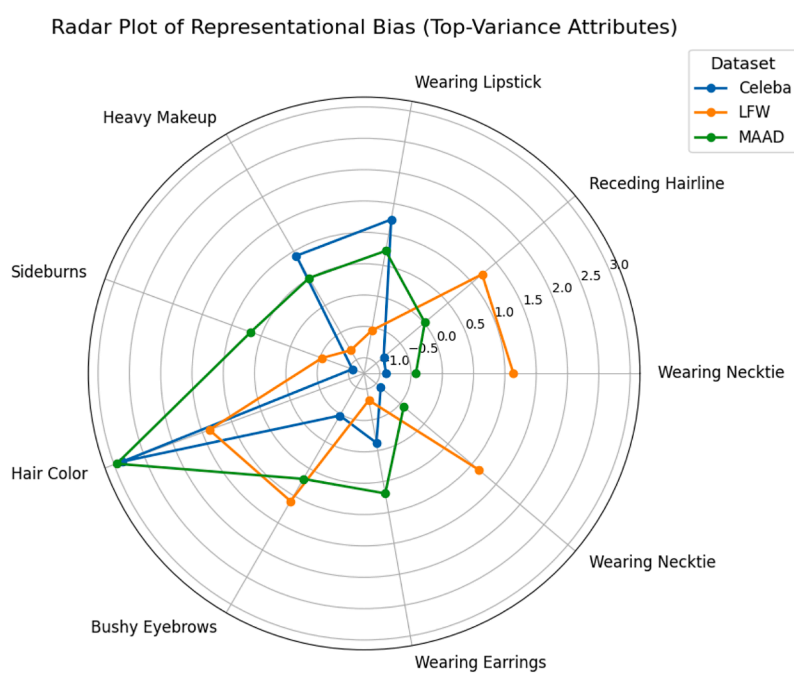

Fig. 3. Radar plot of composite *Z-scores* for the eight visual attributes with highest representational bias variability across datasets, based on averaged normalized values from multiple bias-sensitive metrics.

and conceptual clarity, while minimizing redundancy and simplifying analysis.

5.1.4. Dataset-specific patterns

To examine dataset-specific effects, Fig. 3 reports composite *Z*-scores for the eight visual attributes with the highest inter-dataset variance. These attributes span both appearance-based and gender-expressive traits (e.g., *Wearing Necktie*, *Receding Hairline*, *Wearing Lipstick*). While overall patterns are consistent, notable divergences emerge: *Wearing Necktie* and *Receding Hairline* score higher in CelebA and LFW, likely reflecting differences in gender and age distributions, whereas *Wearing Lipstick* and *Heavy Makeup* appear more balanced or prominent in MAAD. Other traits, such as *Hair Color* and *Bushy Eyebrows*, show moderate variability, potentially influenced by annotation protocols or cultural context. These findings highlight the need for dataset-specific bias evaluations, as attribute distributions vary significantly across datasets. To

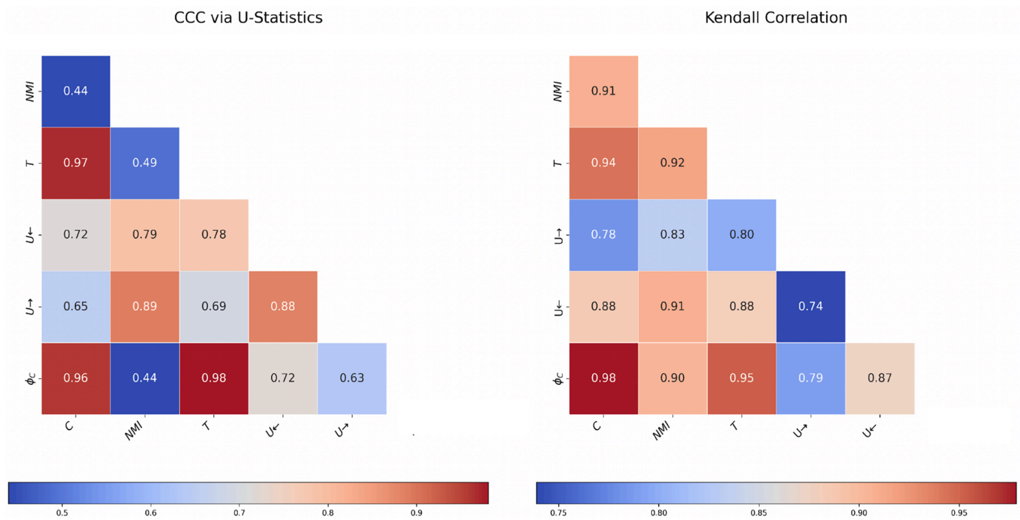

Fig. 4. Heatmaps of Concordance Correlation Coefficient (CCC, left) and Kendall's Tau (right) showing agreement between stereotypical global bias metrics. Dark red indicates strong agreement (values near 1), while blue indicates low agreement (values near 0), as per the color scales below each plot. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

further probe attribute-level disparities, Supplementary Fig. S2 displays individual bias metric values for the soft attribute *Bald*, evaluated with respect to *Gender*, *Age*, and *Race* across CelebA, LFW, and MAAD. The results reveal substantial variability across both metrics and datasets, underscoring the importance of context-aware and group-sensitive auditing strategies.

5.1.5. Key takeaways for practitioners

- Despite conceptual diversity, most representational bias metrics exhibit strong collinearity. A small subset-e.g., *1-BP*, *H*, IR^{-1}, and *SEI*- retains over 98 % of the total variance, supporting the use of compact metric configurations in practice.
- The strong alignment with dominance metrics ($\tau > 0.98$) supports the use of the composite score as a summary indicator, though it should be interpreted in conjunction with metric-level variability.
- Attributes such as *Mustache*, *Goatee*, and *Bald* consistently rank as the most biased across datasets and should be prioritized in mitigation workflows.
- Seemingly stylistic features (e.g., *Wearing Hat*, *Rosy Cheeks*) show consistent cross-dataset bias, suggesting underlying cultural or demographic associations.
- Attributes such as *Wearing Necktie* and *Receding Hairline* exhibit significant variance across datasets, underscoring the necessity of cross-dataset validation. *Hair Color*, while not among the most globally biased attributes, displays high sensitivity to metric configuration and may warrant additional attention during bias assessments.
- Attributes with balanced global scores (e.g., *Age*, *Big Lips*) may still present intersectional bias and should be further analyzed through subgroup analysis.

5.2. Stereotypical global bias metrics

5.2.1. Agreement and correlation among bias metrics

Fig. 4 presents a side-by-side comparison of agreement levels among the evaluated stereotypical bias metrics, based on two complementary statistical measures: the *CCC* on the left and *Kendall's Tau* on the right. This dual representation highlights the difference between numerical agreement and ordinal consistency, offering a nuanced view of how the metrics relate to each other.

The combined results from the CCC and Kendall's Tau analyses provide a detailed understanding of the relationships among the metrics, capturing both absolute concordance and ranking behavior. The metrics ϕ_C, T, and C exhibit particularly strong alignment across both measures, with pairwise CCC values exceeding 0.96 (ϕ_C-T: 0.98, T-C: 0.97, ϕ_C-C: 0.96) and Kendall's Tau values above 0.95. This high level of agreement indicates that these metrics are not only strongly correlated, but also numerically compatible-producing outputs that are nearly interchangeable in both magnitude and relative ordering. Among these relationships, the ϕ_C-T pair is especially notable, with CCCs above 0.95, classified as “strong” and “substantial” by Altman and McBride, respectively [1]. The U-based metrics -$U^{\rightarrow}$ and $U^{\leftarrow}$- show moderate to strong alignment with the core group. $U^{\leftarrow}$, which captures the influence of demographic attributes on soft labels (e.g., the extent to which gender determines the prediction *Wearing Lipstick*), exhibits CCC values ranging from 0.72 to 0.79 and Kendall's Tau between 0.88 and 0.91. These scores reflect strong ordinal consistency and moderate numerical compatibility. $U^{\rightarrow}$, on the other hand, measures the reverse dependency-i.e., how much demographic information can be inferred from soft attribute labels. This directionality highlights cases where the output of a system might leak sensitive information. The metric shows slightly lower CCCs (0.63-0.69) and Kendall's Tau values between 0.78 and 0.83, indicating weaker numerical alignment but still meaningful rank-level agreement. These differences underscore the importance of incorporating directional asymmetry in bias analysis. The U metrics complement symmetric measures like ϕ_C and T, offering better sensitivity to structural imbalances and potential causal pathways-critical in fairness-aware modeling. In contrast, NMI exhibits numerical misalignment (CCC: 0.44-0.49) but high rank consistency (Kendall's Tau > 0.90 with ϕ_C, T, and C). This indicates that while NMI is unsuitable for direct comparisons, it remains valuable for ranking-based evaluations as a secondary diagnostic metric.

5.2.2. Testing for metric interchangeability

While correlation and concordance suggest a degree of similarity, practical applications-particularly those involving fairness-often demand a stricter standard: statistical interchangeability. To assess this, we employed the *TOST* procedure. The test failed to reject the null hypothesis of non-equivalence. This indicates that, despite strong correlations and concordance among several metric pairs, formal statistical equivalence cannot be established. In practical terms, this means that the metrics, though related, may diverge beyond acceptable thresholds for equivalence. As such, they should not be assumed interchangeable in contexts requiring precise numerical comparability. These findings caution against interpreting high correlation as sufficient for metric

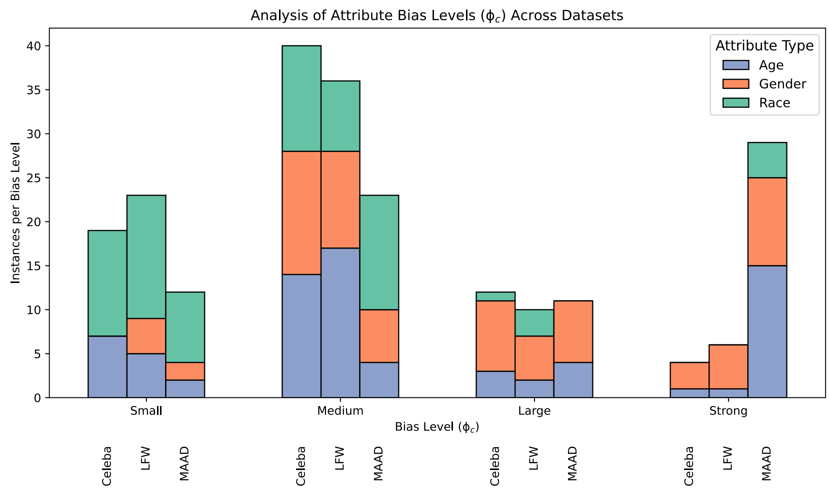

Fig. 5. Distribution of demographic-attribute pairings across bias severity levels, as quantified by ϕ_C.

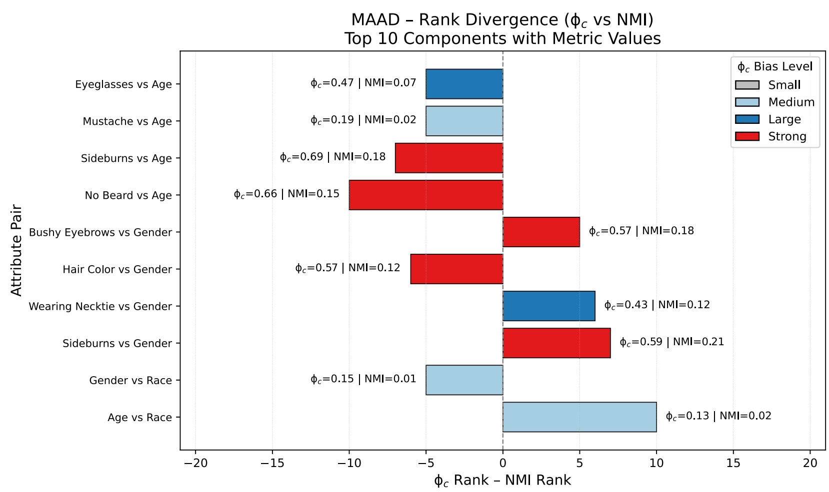

Fig. 6. Top 10 attribute pairs in the MAAD dataset with the largest ranking divergence between ϕ_c and NMI. Bars show the difference in rankings; positive values indicate higher importance under ϕ_c. Colors reflect ϕ_c bias levels (*Small, Medium, Large, Strong*); labels report exact ϕ_c and *NMI* values.

substitution, particularly in fairness-critical analyses where small discrepancies can carry significant implications.

5.2.3. Dataset-level bias intensity via ϕ_C

Among the metrics showing strong mutual agreement, we prioritize the analysis of ϕ_C due to its documented interpretive thresholds in the literature, which enable consistent and standardized bias evaluation. As shown in Fig. 5, MAAD exhibits the highest concentration of attribute pairs classified under the *Strong* bias category, with an average ϕ_C of 0.40 ±0.28-substantially higher than the corresponding values in CelebA (0.21±0.15) and LFW (0.20± 0.16). This trend suggests that MAAD encodes stronger and more systematically polarized attribute dependencies. By contrast, CelebA and LFW display lower and more uniformly distributed bias levels. Despite differences in magnitude, certain associations-such as *Wearing Lipstick* vs *Gender* and *Wearing Earrings* vs *Gender*-consistently yield high ϕ_C values across all datasets, indicating entrenched stereotypical correlations, particularly around gendered traits. Other associations appear dataset-specific. For example, the link between *Big Nose* and *Race* is classified as *Strong* only in MAAD, possibly due to racial imbalance or dataset-specific labeling conventions. Frequency analysis supports these patterns: in MAAD, fewer than 0.05 % of male samples are labeled with *Wearing Lipstick*, compared to over 34 % of female samples, yielding a ϕ_C of approximately 0.88. Likewise, *Gray Hair* is predominantly assigned to the senior group (6.92 %), reflecting a biologically plausible yet potentially reductive pattern if used naively in predictive tasks.

These findings highlight the importance of supplementing quantitative metrics such as ϕ_C with qualitative data inspections. Such a combined approach is essential to distinguish biases stemming from social stereotypes, annotation inconsistencies, or structural imbalances-each calling for tailored mitigation strategies based on the specific modeling context.

5.2.4. Metric disagreement: ϕ_C vs NMI rankings

To investigate inconsistencies across bias metrics, we examined the relative rankings produced for individual attribute pairs. Fig. 6 highlights the top 10 components in MAAD exhibiting the largest ranking discrepancies between ϕ_C and NMI. This analysis illustrates how distinct metrics may produce divergent assessments of component importance. For example, pairs such as *Eyeglasses vs Age* and *Mustache vs Age* receive high ϕ_C scores but are deprioritized by NMI, revealing scale-related misalignment. Conversely, pairs like *Sideburns vs Gender* and *Gender vs Race* are ranked higher by NMI despite lower ϕ_C values, suggesting NMI may emphasize ordinal relationships over absolute association strength. Each bar in the figure is color-coded based on ϕ_C bias level, offering insight into whether discrepancies emerge primarily in low, moderate, or strong bias contexts. Notably, even pairs with high ϕ_C values sometimes display substantial disagreement, indicating that components with evident stereotypical associations may be interpreted differently depending on the chosen metric.

These results reinforce the need to employ multiple, complementary bias metrics. Large divergences highlight ambiguous or multidimensional relationships between attributes, where relying on a single measure could lead to misinterpretations-either underestimating or overstating the presence of bias. A metric-aware approach is therefore essential for reliably identifying the most critical attribute pairings. Beyond symmetric associations, the following section explores directional dependencies to better understand the causal influence of demographic traits on facial attributes.

5.2.5. *Directional asymmetry in attribute dependencies*

To deepen the global analysis of bias metrics, we examine directional asymmetries in attribute associations via $\Delta U = U^{\leftarrow} - U^{\rightarrow}$, distinguishing whether a demographic attribute influences a visual feature or vice versa. This distinction is crucial for identifying whether models internalize societal biases (*stereotypical imprinting*, high $U^{\leftarrow}$) or expose sensitive traits through inference (*demographic leakage*, high $U^{\rightarrow}$).

As shown in Fig. 7, and detailed in the Supplementary Material, each dataset exhibits a distinct asymmetry profile. MAAD shows the strongest directional divergences, with high positive ΔU values for attribute pairs such as *Hair Color vs Race* and *Hair Color vs Gender*, and negative extremes for *Double Chin vs Age* and *Rosy Cheeks vs Gender*, pointing respectively to stereotypical conditioning and demographic inference. LFW, by contrast, displays more moderate and mostly negative asymmetries, particularly for gendered features such as *Wearing Necktie* or *No Beard*, which suggest demographic leakage with limited stereotyping. CelebA reflects similar patterns but with more pronounced gender asymmetries, and also exhibits notable negative ΔU values for age-related traits such as *Chubby* and *Bald*, revealing the presence of visual proxies for sensitive traits even in benchmark datasets. A cross-dataset comparison reveals that only two attribute pairs—*Hair Color vs Race* and *Hair Color vs Gender*—consistently appear among the top asymmetric components across all three datasets, suggesting deeply entrenched associations. Age-related asymmetries are more dataset-specific, shared primarily between MAAD and CelebA, while gender-related ones dominate in LFW and CelebA. Crucially, these directional asymmetries align with empirical co-occurrence patterns. For instance, in MAAD, over 98 % of samples labeled as *Blond Hair* belong to White individuals, and only 0.04 % are male—explaining the high $U^{\leftarrow}$ scores. See Supplementary Material for a comparative view supporting these asymmetries. Similar trends are observed in both CelebA and LFW, where visual traits such as hair color and facial hair exhibit strong correlations with race and gender-highlighting systemic distributional imbalances rather than mere modeling artifacts. These findings emphasize the need for mitigation strategies that account for the directionality of bias. High $U^{\rightarrow}$ values (e.g., *Wearing Lipstick vs Gender*), suggesting the adoption of privacy-preserving representations to prevent demographic inference. Conversely, high $U^{\leftarrow}$ values call for *fairness-aware debiasing* to reduce reliance on demographic information when predicting visual traits. Both scenarios require not only technical

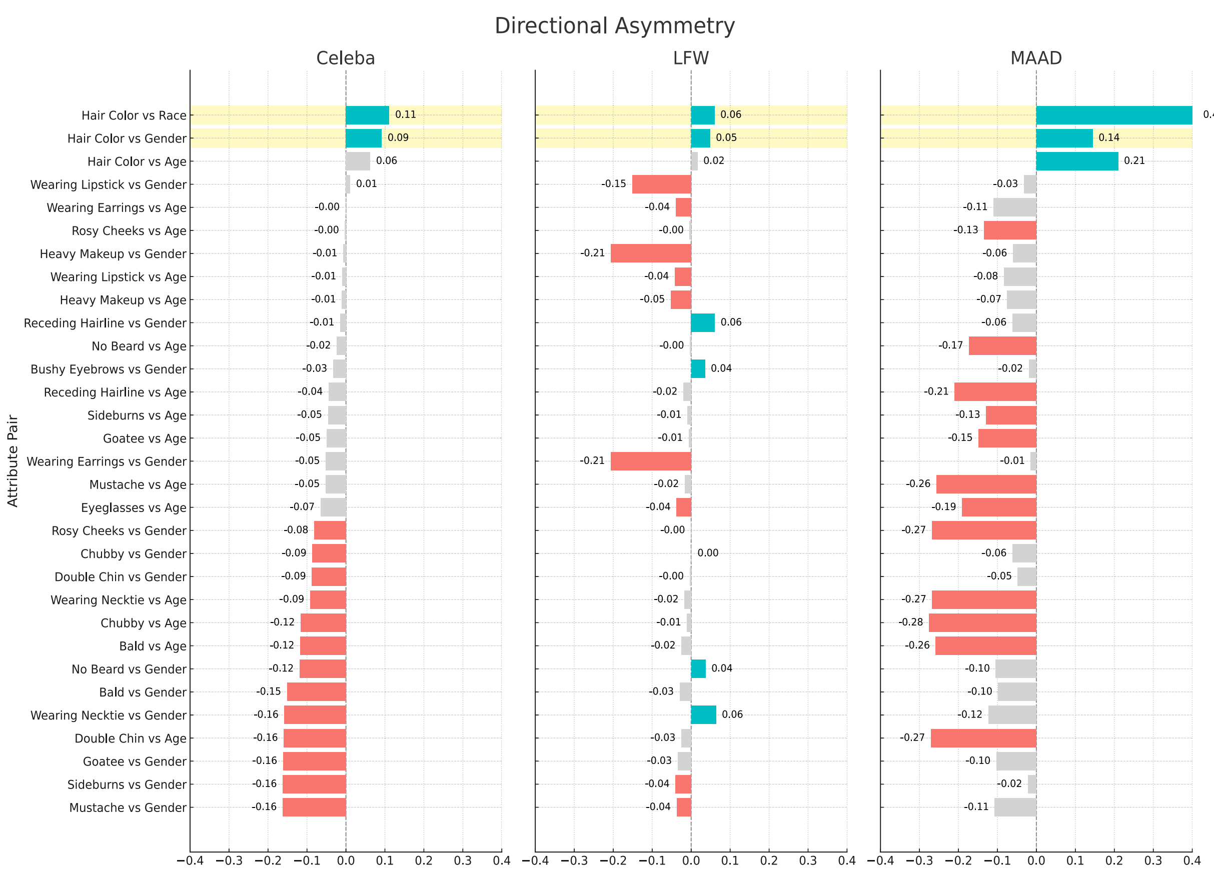

Fig. 7. Directional asymmetry ($U^{\leftarrow} - U^{\rightarrow}$) for selected attribute pairs across datasets. Positive values (turquoise) indicate demographic-to-soft-attribute influence; negative values (coral) indicate soft-attribute-to-demographic influence. The x-axis is aligned for cross-dataset comparability; shaded areas mark the most asymmetric pairs.

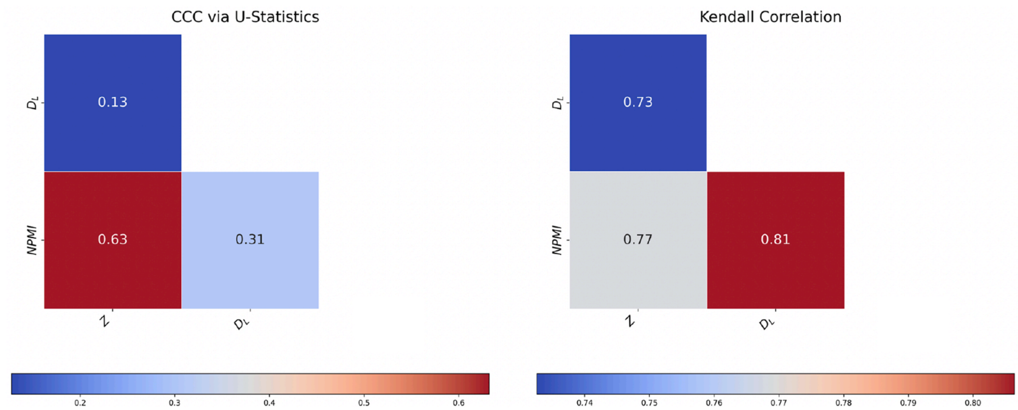

Fig. 8. Heatmaps of Concordance Correlation Coefficient (CCC, left) and Kendall's Tau (right) showing agreement between stereotypical local bias metrics. Dark red indicates strong agreement (values near 1), blue indicates low agreement (values near 0), based on the color scales below each plot. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

intervention, but also transparent reporting of bias pathways and dataset composition.

5.2.6. Key takeaways for practitioners

A summary of cross-dataset bias intensity and metric behavior is provided below, with emphasis on key differences in structural bias profiles.

- ϕ_C, T, and C exhibit high numerical and ordinal agreement (CCC > 0.96, $\tau > 0.95$) and can be used interchangeably for ranking bias severity.
- MAAD reveals deeper structural bias, with higher average ϕ_C and a greater number of strongly biased pairs-often reflecting intersectional dependencies involving both race and gender.
- In contrast, CelebA and LFW exhibit predominantly gender-based patterns, with less severe but more consistent asymmetries.
- Metric disagreement, especially at the rank level (e.g., ϕ_C vs NMI), highlights the presence of complex or non-linear associations that single metrics may fail to capture.
- Recurrent patterns across datasets-such as *Hair Color vs Race*-indicate systemic issues in data representation and annotation practices.
- Effective bias mitigation requires metric-aware strategies, as both the strength and nature of bias depend on dataset structure and metric sensitivity.
- TOST results indicate that metrics should not be treated as numerically interchangeable, even when correlated. Use them with awareness of magnitude differences.
- When prioritizing global metric selection, ϕ_C is recommended due to standard interpretability thresholds and broad agreement.
- Directional asymmetries (ΔU) expose distinct bias mechanisms-stereotypical imprinting vs demographic leakage-requiring tailored responses.

5.3. Stereotypical local bias metrics

5.3.1. Agreement and correlation among bias metrics

Fig. 8 shows pairwise agreement among local bias metrics, evaluated via *CCC* and *Kendall's Tau*. Kendall's τ values are consistently high ($[0.73, 0.81]$), indicating strong alignment in the ordinal ranking of group-attribute associations. However, CCC reveals notable divergence in value-level agreement, with $NPMI$ and Z showing moderate concordance (CCC = 0.63), while D_L exhibits low alignment with Z (CCC = 0.13). This suggests that D_L is more sensitive to absolute deviations in association strength, whereas Z and $NPMI$ better preserve rank-order similarity.

5.3.2. Metric-specific patterns of demographic bias

Following the metric agreement analysis, we examine which demographic dimensions-*Age*, *Gender*, or *Race*-most frequently appear in the top biased group-attribute pairs per metric. This reveals how the choice of metric and the underlying data distribution influence which forms of bias are most prominent. Fig. 9 reports the relative contributions of each dimension within the top 20 % most biased associations. CelebA displays a persistent gender-driven pattern, with gender accounting for over 50 % of high-bias cases in $NPMI$, and remaining dominant in Z (0.39) and D_L (0.49). In contrast, *Age* and *Race* contribute far less, generally remaining below 0.30-a reflection of the dataset's highly gender-coded visual cues. LFW exhibits a more balanced profile: while gender remains prominent (e.g., 0.54 in $NPMI$), *Age* and *Race* gain importance, particularly in D_L and Z (0.39 and 0.34, respectively). MAAD reveals a distinct dynamic: *Age* overwhelmingly dominates across all metrics (e.g., 0.58 in D_L), despite the dataset's balanced design. This suggests that broad or uneven age distributions may elicit stronger metric responses. *Gender* retains a secondary role (0.29–0.39), while *Race* contributes minimally. These trends align with Fig. 5, confirming that CelebA exhibits widespread and severe gender-related bias, while MAAD concentrates fewer but more extreme, age-related cases. Such localized disparities may not emerge from marginal statistics alone (Table 4, Supplementary Material), underscoring how stereotypical bias often arises from interactions rather than isolated distributions. Violin plots in Fig. 10 reinforce this: CelebA's peaked profiles point to concentrated bias (mostly gender), LFW shows broader dispersion, and MAAD displays heavier tails-especially in Z-reflecting strong sensitivity to demographic interactions.

5.3.3. Attribute-specific insights

Aggregated Demographic Bias Analysis. To identify soft visual attributes that systematically contribute to demographic bias, an aggregated composite analysis was performed across all datasets and metrics. For each combination of soft attribute and demographic group, we computed a composite bias score by averaging normalized values (*Z-scores*) across three selected metrics. This global analysis aims to reveal overarching bias patterns associated with demographic groups, independently of any specific dataset. As reported in Table 1, bias metrics were computed for all soft attribute-demographic variable pairs. The absolute values of these scores were then aggregated by demographic group to estimate the overall bias intensity associated with each dimension. This aggregation enables a comparison of each demographic group's tendency to co-occur with biased attribute patterns, regardless of which attributes are involved. The results indicate that *Gender* exhibits the highest average bias across all metrics, followed by *Age* and then *Race*. Interestingly, D_L provides overall more conservative estimates, especially in race-related combinations.

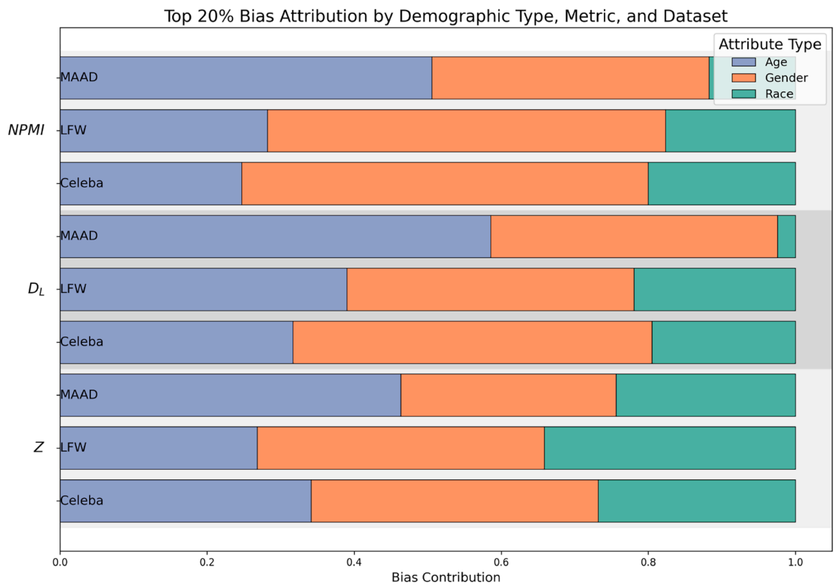

Fig. 9. Normalized distribution of demographic attributes among the top 20 % most biased pairs, ranked by $NPMI$, D_L, and Z across datasets.

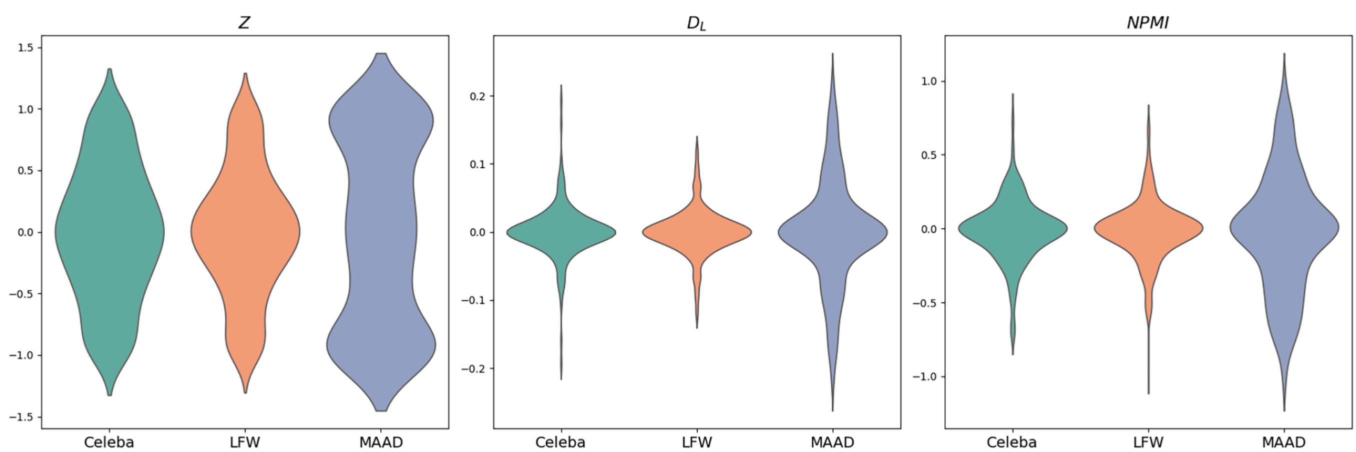

Fig. 10. Violin plots showing the distribution of Z, D_L, and $NPMI$ metric contributions across datasets.

Table 1
Summary of absolute bias scores (mean ± standard deviation) by demographic category and metric.

Group	Z	D_L	$NPMI$
Age	0.473 ± 0.330	0.030 ± 0.037	0.192 ± 0.205
Gender	**0.712** ± 0.310	**0.062** ± 0.053	**0.284** ± 0.227
Race	0.395 ± 0.316	0.012 ± 0.015	0.115 ± 0.148

Since the metrics differ in scale and statistical behavior, we avoided the direct aggregation of raw scores. Instead, we computed a composite bias score by averaging the absolute bias values previously normalized via Z-scores. This approach enables meaningful comparisons across metrics while preserving their individual contributions Fig. 11. Among the top-ranked soft attributes, we observe a clear overrepresentation of stylized or gender-coded traits, such as *Heavy Makeup*, *Wearing Lipstick*, and *Wearing Earrings*, particularly in combination with $Gender$. This supports the observation that visual attributes typically associated with gender tend to exhibit high demographic imbalance within datasets, even in the absence of an explicit gender label. Interestingly, the case of *Hair Color* is more nuanced: it does not always induce strong positive or negative bias on its own. However, more granular analysis reveals that in specific combinations-such as *Blond Hair* + *Female* or *Gray Hair* + *Senior*-its contribution to overall bias becomes strongly significant. This suggests that *Hair Color* may contribute to demographic bias in interaction with other traits, suggesting a potential modulating effect that warrants further investigation. Conversely, *Wearing Hat* consistently receives low composite scores and low standard deviation, particularly when evaluated against $Race$. This indicates that it does not systematically contribute to demographic bias in the datasets analyzed and can be considered a visually neutral attribute.

Dataset-specific breakdown. To assess the generalizability of bias patterns, we performed a per-dataset analysis by computing the mean absolute bias scores of the top-ranked soft attributes. This enabled the distinction between (i) *consistent bias across datasets*, indicating structural associations with demographic groups, and (ii) *dataset-specific bias*, likely driven by contextual or annotation-related factors. The analysis focused on the three most biased soft attributes per demographic group, as ranked by composite scores. For *Age*, the most consistently bias-prone attributes were *Wearing Lipstick*, *Heavy Makeup*, and *Big Nose*; for *Gender*, *Wearing Lipstick*, *Heavy Makeup*, and *Wearing Earrings*; and for *Race*, *Pointy Nose*, *Hair Color*, and *Wavy Hair*. These attributes emerge as primary candidates for bias monitoring, mitigation, and interpretability efforts across datasets. These attributes span both stylistic features (e.g. makeup and lipstick) and morphological traits (e.g., nose shape),

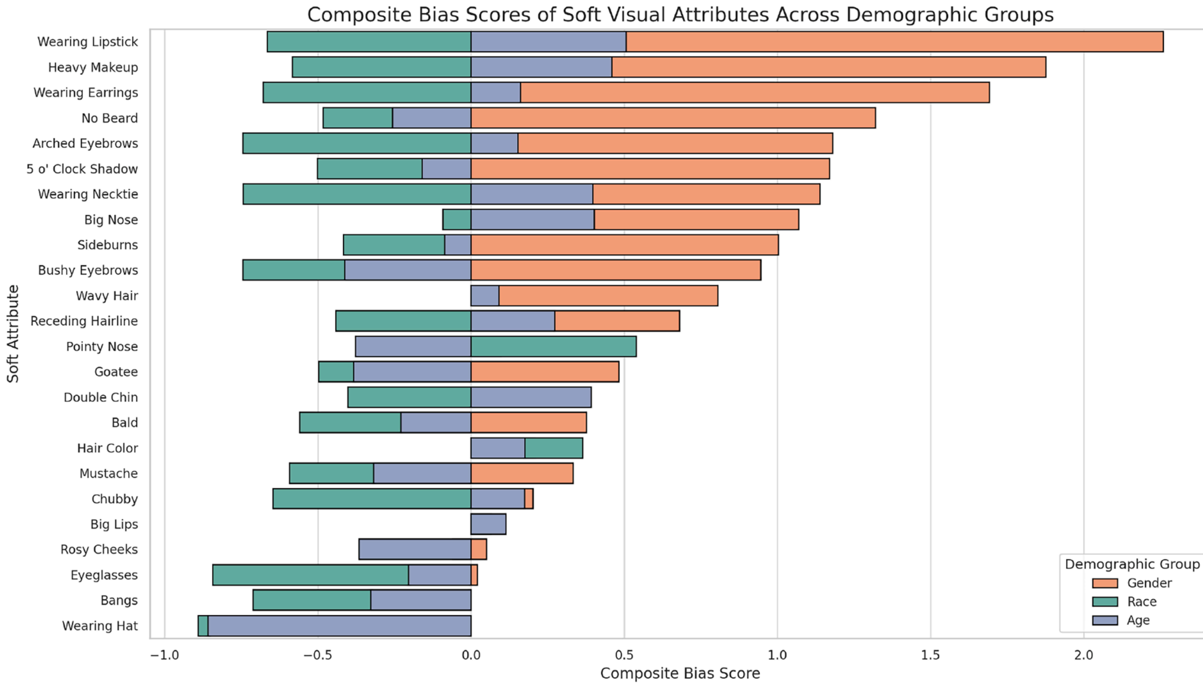

Fig. 11. Composite bias scores of soft visual attributes across demographic groups. Each bar represents the normalized bias intensity of a soft attribute, averaged across the three metrics Z, D_L, and $NPMI$.

suggesting that both aesthetic conventions and facial structure can act as demographic markers within visual data. When examining the *age*-related attributes, we observe that all three show significantly higher bias scores in the MAAD dataset compared to CelebA and LFW. However, *Wearing Lipstick* and *Heavy Makeup* display more consistent levels of severity across the three datasets, whereas *Big Nose* appears disproportionately influential in MAAD, likely contributing substantially to its overall composite score for this group. For *Gender*, both MAAD and LFW concur on the presence of strong bias associated with *Wearing Earrings*, while CelebA diverges with lower scores for this attribute. In contrast, there is high agreement across all three datasets for the attributes *Wearing Lipstick* and *Heavy Makeup*, reinforcing their status as visually gender-coded features prone to imbalance. Finally, in the case of *Race*, we observe full alignment across all three datasets: the attributes *Pointy Nose*, *Hair Color*, and *Wavy Hair* consistently rank among the most biased, indicating a shared distributional skew that transcends dataset-specific characteristics. These patterns reinforce that demographic bias does not always manifest uniformly across datasets or metrics.

Metric-Level Disagreement. To evaluate metric consistency, we computed the *inter-metric standard deviation* for each soft attribute-demographic group pair. This value reflects the variability among the three normalized bias scores, serving as an indicator of *metric disagreement*. Prior to this, all absolute bias values were *Z-score* normalized within each metric to mitigate scale and dispersion differences. Higher standard deviation indicates greater disagreement among metrics. Ranking the combinations by this measure allowed us to identify the five most unstable attribute-demographic pairs. As illustrated in Fig. 12, the top-ranked combination, *Mustache + Gender*, shows a large positive score from Z (1.39), a negative score from D_L (-0.43), and a near-zero $NPMI$ (0.04), leading to the highest standard deviation (0.95). In this case, the metrics diverge due to extreme asymmetry: *Mustache* is almost exclusive to male subjects. D_L returns a high value, as it captures the raw absolute difference between observed and expected joint probabilities, regardless of overall frequency. Z yields a negative score, reflecting underrepresentation relative to the maximum possible deviation given the marginals. $NPMI$ remains close to zero because, although the co-

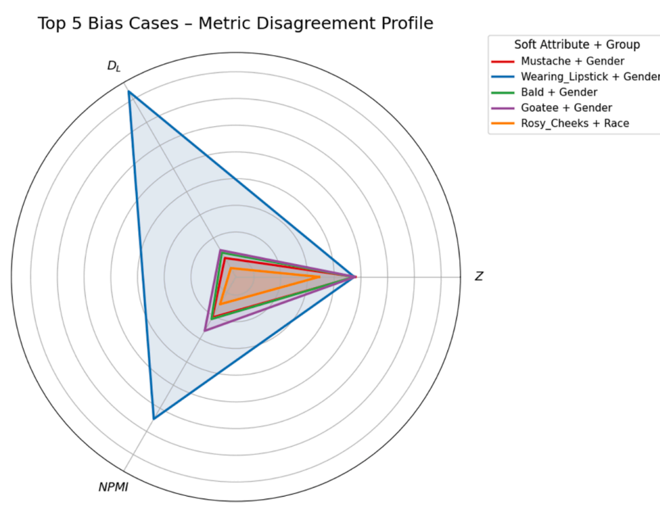

Fig. 12. Radar plot illustrating metric-level disagreement for the five most unstable combinations.

occurrence is extremely rare, it closely matches what would be expected under independence-resulting in little to no informational deviation. Other combinations-such as *Bald + Gender*, *Goatee + Gender*, and *Wearing Lipstick + Gender*-also exhibit significant variability across metrics, reflecting differences in how the metrics respond to categorical imbalance, co-occurrence frequency, and marginal distribution. Interestingly, *Rosy Cheeks + Race* appears among the top five, indicating that metric disagreement is not limited to gender-coded traits. In this case, D_L again deviates from the other two metrics, further highlighting its distinct sensitivity to certain forms of distributional divergence.

5.3.4. Key takeaways for practitioners

- Metrics are not interchangeable. Each bias metric captures different statistical properties (e.g., co-occurrence, marginal imbalance, rank distribution) and may lead to divergent conclusions if interpreted in isolation.

- Metric disagreement is informative. High variability across metrics can uncover hidden or conditional bias patterns that composite or single-metric analyses might obscure.
- Composite scores improve interpretability but require validation. While composite scores offer a useful summary, they should always be interpreted in conjunction with inter-metric agreement checks to avoid misleading conclusions.
- Bias is often conditional, not global. Many high-bias combinations emerge from specific interactions between visual features and demographic groups, rather than overall frequency imbalances.
- Stylistic attributes are frequent bias carriers. Features like *Heavy Makeup* and *Wearing Lipstick* are highly associated with demographic groups-even when those groups are not explicitly labeled-indicating the presence of latent visual stereotypes.
- Fairness pipelines should include metric-level agreement checks. Practitioners are encouraged to define thresholds for inter-metric variability to flag unstable combinations and assess the robustness of bias estimates.
- Visual stereotypes may reinforce social norms. Some of the most biased attributes reflect strong cultural or aesthetic connotations, suggesting that demographic bias can encode societal expectations beyond statistical asymmetry.

6. Conclusions

This work presented a unified framework to analyze demographic bias in facial attribute datasets, integrating representational and stereotypical perspectives through diverse statistical metrics. Our results show that bias is multifaceted, emerging from complex, attribute- and dataset-specific patterns that cannot be captured by single metrics or global summaries. These reflect structural imbalances and culturally encoded stereotypes. Through ordinal and numerical agreement analyses, we identified metric families providing either redundant or complementary insights. Symmetric measures (e.g., ϕ_C, T, C) showed strong internal consistency and support ranking-based evaluations, while directional and information-theoretic metrics revealed critical asymmetries, pointing to risks of stereotype reinforcement and demographic leakage. To enable robust cross-metric comparisons, we adopted a composite scoring approach based on normalized outputs. This helped detect both recurrent bias configurations-especially those linked to gender-coded traits-and subtler imbalances tied to dataset composition and demographic interactions. Overall, our findings reinforce the importance of adopting multi-metric, multi-level evaluations when assessing bias in visual datasets, especially in fairness-critical applications. Such understanding is crucial not only for fairness auditing but also for designing mitigation strategies that respect the sociocultural context of facial recognition systems. As future work, we plan to systematically assess the impact of dataset-level bias on the performance of soft biometric recognition models. We will conduct controlled experiments using both original and bias-corrected datasets to quantify how annotation refinements and demographic rebalancing affect model accuracy and fairness. Additionally, we aim to explore the use of generative models-such as diffusion- and GAN-based architectures-to synthesize demographically balanced samples. This synthetic rebalancing will support targeted model retraining and help evaluate its effectiveness in mitigating the amplification of harmful stereotypes. Through these efforts, we move from bias characterization toward a principled evaluation of corrective interventions at the model level.

CRediT authorship contribution statement

Lucia Cascone: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Michele Nappi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Chiara Pero:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Xinggang Wang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.patcog.2025.112416

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